

ES666 Computer Vision

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Imaging · Linear algebra & SVD · Convolution & Fourier · Filtering · Edges & corners

Scale space & SIFT · Projective geometry · Homography, DLT & RANSAC · Cameras

Epipolar geometry (F & E) · Multi-view stereo & SfM · Optical flow · Photometric stereo and inverse rendering

Eighteen hours of teaching a computer to do what your retina does for free — with proofs, because we don't trust nice pictures.

Course outline — eighteen hours, four parts

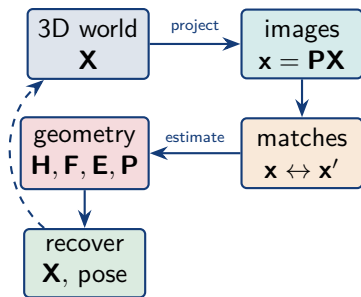
Part I — Foundations (Hr 1–3): image formation; the SVD; least squares.

Part II — Filtering (Hr 4–6): convolution; Fourier; smoothing.

Part III — Features (Hr 7–10): edges; Canny/Harris; scale space; SIFT/SURF/ORB.

Part IV — Geometry, motion & light (Hr 11–18): projective geometry; homography; DLT/RANSAC; camera calibration; epipolar geometry; multi-view stereo & SfM; optical flow; photometric stereo.

The big picture: geometric vision



Forward: the world projects to images (Hr 14). **Inverse:** from images and matches we estimate geometry (Hr 11–15) and recover structure & motion (Hr 15–16). Part IV is this inverse problem.

Part I Roadmap

1 — Introduction & image formation

CV vs. image processing / graphics / generative AI; image as a function; sampling, quantization & Nyquist; why CV is hard; **pinhole model & perspective projection**; lens & depth of field; the sensing pipeline; demosaicing, gamma, CRF.

2 — Linear algebra I (with proofs)

Four subspaces; spectral theorem; **SVD existence proof**; eigendecomposition link; rank/range/null space, norms, det, κ (all proved); worked 2×2 SVD; **Eckart–Young proof**.

3 — Linear algebra II (with proofs)

Normal-equations derivation; residual orthogonality; projection matrix; pseudo-inverse via SVD; **homogeneous $Ax = 0$ proof**; minimum-norm (Lagrange); regression; total least squares; conditioning & normalization.

Why this first?

Every later topic reduces to *solving a linear system* or *minimizing a residual*. Master this and the rest is geometry.

Sources: Szeliski (2022); Hartley & Zisserman (2004); Trefethen & Bau (1997).

Part II roadmap (Hours 4–6 of 18)

Hour 4 — Convolution & LTI

Sifting; **LTI** $\Rightarrow$ **convolution** (proof); commutativity / associativity / length (proofs); FIR/IIR; **BIBO stability** proof; correlation & NCC; 2D **separability** proof; Toeplitz/circulant view.

Hour 5 — Fourier analysis

Complex exponentials as **eigenfunctions** (proof); CTFT/DTFT/DFT/FFT; **convolution theorem** (proof); shift/scaling/derivative (proofs); circular convolution; 2D separability; **Gaussian self-duality** (proof); sampling & aliasing.

Hour 6 — Image filtering

Box filter (sinc response); **separability & integral images** (proofs); Gaussian **separability + semigroup** (proofs); binomial/CLT; derivative-of-Gaussian; **noise-variance proof**; median / bilateral / NL-means; anti-aliased pyramids; sharpening.

Thread

One idea runs through all three hours: *convolution in space = multiplication in frequency*. Hour 4 builds it, Hour 5 explains it, Hour 6 exploits it.

Sources: Oppenheim & Schaffer; Bracewell; Gonzalez & Woods (2018); Szeliski (2022).

Part III roadmap (Hours 7–10 of 18)

Hour 7 — Gradients, edges, LoG, DoG

Finite differences (accuracy proof); Sobel separability; noise/derivative trade-off; **LoG closed form**; heat equation; **DoG $\approx$ LoG** proof; structure tensor; Hough.

Hour 9 — Scale space & blobs

Scale-space axioms; **diffusion-equation proof**; pyramids & reconstruction; **scale normalization** & characteristic-scale proof; DoG scale space; 26-neighbour extrema.

Hour 8 — Canny & Harris

Canny criteria & optimal detector; NMS; hysteresis; **auto-correlation** $\rightarrow$ **structure tensor** derivation; eigenvalue classification; $R = \det \mathbf{M} - k \operatorname{tr}^2 \mathbf{M}$; **rotation-invariance proof**.

Hour 10 — SIFT / SURF / ORB

DoG keypoints; **sub-pixel & edge-rejection** proofs; orientation; 128-D descriptor; normalization; **ratio test**; SURF; FAST/BRIEF/ORB; **RANSAC** count proof.

Sources: Marr & Hildreth (1980); Canny (1986); Harris & Stephens (1988); Lindeberg (1994); Lowe (2004).

Part IV roadmap

Geometry of one & two views

- Hr 11** Projective geometry of the plane
- Hr 12** Transformations & the homography
- Hr 13** DLT & robust estimation (RANSAC)
- Hr 14** Camera models & calibration

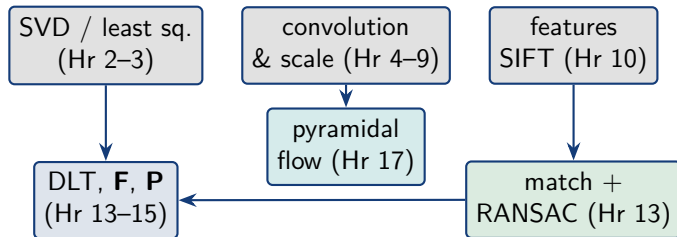
Two views, reconstruction, motion & light

- Hr 15** Epipolar geometry: F & E
- Hr 16** Multi-view stereo & SfM
- Hr 17** Optical flow + coarse synthesis
- Hr 18** Photometric stereo & shape

The arc: *represent* geometry (11) → *estimate* it (12–13) → *calibrate* cameras (14) → *relate* two views (15) → *reconstruct* scenes (16) → *track* motion (17) → *recover shape from light* (18).

Sources: Hartley & Zisserman (2004); Szeliski (2022); Forsyth & Ponce (2012).

How Part IV builds on Parts I–III



Every tool from the first three parts is load-bearing here: SVD solves the homogeneous systems, scale-space feeds coarse-to-fine flow, and SIFT supplies the correspondences that geometry consumes.

Prerequisites in one slide

Four things from earlier parts to keep within arm's reach:

- ▶ **SVD** (Hr 2–3): $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$; smallest singular vector solves $\min_{\|\mathbf{h}\|=1} \|\mathbf{A}\mathbf{h}\|$; Eckart–Young gives best low-rank.
- ▶ **Eigenvalues** & the **structure tensor** (Hr 8): corner strength from λ_1, λ_2 of $\mathbf{M}$.
- ▶ **Scale space & pyramids** (Hr 9): Gaussian/Laplacian pyramids, coarse-to-fine.
- ▶ **Features** (Hr 10): SIFT detection & matching, the ratio test.

If a step ever feels like it fell from the sky, it almost certainly traces back to one of these four. Look up, not out the window.

Notation & conventions

- ▶ scalars a ; vectors $\mathbf{x}$ (bold, columns); matrices $\mathbf{A}$
- ▶ $\mathbf{A}^\top$ transpose; $\mathbf{A}^{-1}$ inverse; $\mathbf{A}^+$ pseudo-inverse
- ▶ $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\top \mathbf{x}}$; $\|\mathbf{A}\|_2, \|\mathbf{A}\|_F$ matrix norms
- ▶ $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$; $\mathbf{x} \perp \mathbf{y} \iff \mathbf{x}^\top \mathbf{y} = 0$
- ▶ $\mathbf{I}$ identity; $\mathbf{0}$ zero vector; $\mathbf{e}_i$ standard basis
- ▶ Col, Null, rank, span, tr, diag as named
- ▶ σ_i singular values; λ_i eigenvalues
- ▶ “ $\simeq$ ” equality *up to non-zero scale* (homogeneous)
- ▶ $I(i, j)$ image intensity; $\mathbf{X} \in \mathbb{R}^3$ scene point; $\mathbf{x} \in \mathbb{R}^2$ image point

Real talk

Grey footnotes list each slide's **primary sources**. Boxed results are the ones to memorize; *Proof* boxes are self-contained.

Notation & conventions (signals edition)

- ▶ $x[n], h[n]$: discrete signals; $x(t)$: continuous
- ▶ $\delta[n]$ Kronecker, $\delta(t)$ Dirac (impulse)
- ▶ $*$ convolution; $\circledast$ *circular* convolution; $\star$ correlation
- ▶ h : impulse response; $H(\omega)$: frequency response
- ▶ $\mathcal{F}\{\cdot\}$ Fourier transform; $X(\omega)$ spectrum
- ▶ DFT, FFT: discrete / fast Fourier transform
- ▶ $\hat{f}$: transform of f ; $|X|, \angle X$: magnitude, phase
- ▶ G_σ : Gaussian of std. dev. σ ; $\text{sinc}(x) = \sin x / x$
- ▶ $I[m, n]$: image; k : kernel size; N : pixel count

Real talk

Grey footnotes give each slide's **primary sources**; boxed results are the keepers; *Proof* boxes are self-contained and end with ■.

Notation & conventions (features edition)

- ▶ $I(x, y)$: image; I_x, I_y : partial derivatives
- ▶ $\nabla I = (I_x, I_y)$: gradient; $\nabla^2 I$: Laplacian
- ▶ G_σ : Gaussian, scale σ ; G'_σ : derivative of Gaussian
- ▶ $L(x, y, \sigma) = G_\sigma * I$: scale space
- ▶ $\text{LoG} = \nabla^2 G_\sigma$; $\text{DoG} = G_{k\sigma} - G_\sigma$
- ▶ $\mathbf{M}$: structure / second-moment tensor; λ_1, λ_2 its eigenvalues
- ▶ $\mathcal{H}$: Hessian; R : Harris response
- ▶ $\mathbf{d}$: descriptor vector; θ : orientation
- ▶ NMS: non-maximum suppression

Real talk

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Notation I — projective geometry

$\mathbf{x} = (x, y, 1)^\top$	homogeneous image point
$\ell = (a, b, c)^\top$	line; incidence $\ell^\top \mathbf{x} = 0$
$\mathbf{x}_1 \times \mathbf{x}_2$	join (line through two points)
$\ell_1 \times \ell_2$	meet (intersection of two lines)
$\mathbb{P}^2, \ell_\infty$	projective plane, line at infinity
$\mathbf{H} (3 \times 3)$	homography; $\mathbf{x}' = \mathbf{H}\mathbf{x}$, 8 DOF
$\mathbf{C}, \mathbf{C}' = \mathbf{H}^{-\top} \mathbf{C} \mathbf{H}^{-1}$	conic and its transform
$\text{Cross}(\cdot)$	cross-ratio (projective invariant)

Notation II — cameras & two views

$\mathbf{P} = \mathbf{K}[\mathbf{R} \mid \mathbf{t}]$	3×4 camera, 11 DOF
$\mathbf{K}$	intrinsics (f_x, f_y, c_x, c_y, s)
$\mathbf{R}, \mathbf{t}, \tilde{\mathbf{c}} = -\mathbf{R}^\top \mathbf{t}$	extrinsics, camera centre
$\boldsymbol{\omega} = \mathbf{K}^{-\top} \mathbf{K}^{-1}$	image of the absolute conic
$\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{R}$	essential matrix (calibrated)
$\mathbf{F} = \mathbf{K}'^{-\top} \mathbf{E} \mathbf{K}^{-1}$	fundamental matrix
$\mathbf{e}, \mathbf{e}'$	epipoles = $\text{Null } \mathbf{F}, \text{Null } \mathbf{F}^\top$
$\ell' = \mathbf{F} \mathbf{x}$	epipolar line of $\mathbf{x}$

Sources: Hartley & Zisserman (2004), Ch. 6, 9.

Notation III — estimation & motion

$\mathbf{A}\mathbf{h} = \mathbf{0}$	homogeneous DLT system
$\mathbf{h}$ = smallest sing. vec.	DLT solution (SVD)
$\mathbf{T}, \mathbf{T}'$	Hartley normalizing transforms
$N \geq \frac{\log(1-p)}{\log(1-w^s)}$	RANSAC iterations
$\mathcal{E}_{\text{Samp}}$	Sampson (geometric) error
(u, v)	optical flow at a pixel
$I_x u + I_y v + I_t = 0$	flow constraint (OFCE)
$\mathbf{M} = \sum \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$	structure tensor (Harris = LK)

Sources: Hartley & Zisserman (2004), §4; Lucas & Kanade (1981).

How to survive these slides

Proof

Self-contained derivations, ending in ■. If you black out mid-proof, that little square is where you come to.

Boxed result

Memorize these. Pull one out and the entire field quietly collapses back into a pile of pixels.

Real talk

The honest asides: intuition, caveats, and whatever the formula politely declined to mention.

Slides marked “Worked” contain real numbers — we did the arithmetic so you can nod as if you would have. The grey text at the bottom credits the people who worked this out decades before any of us showed up.

Sources: Deck conventions.